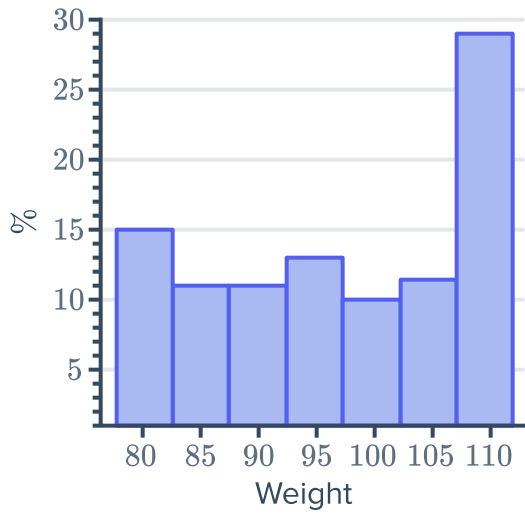


The normal distribution

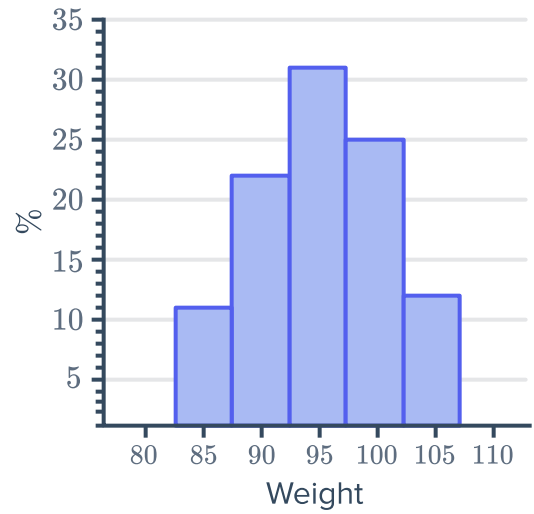


1 State whether the following graphs of data are normally distributed:

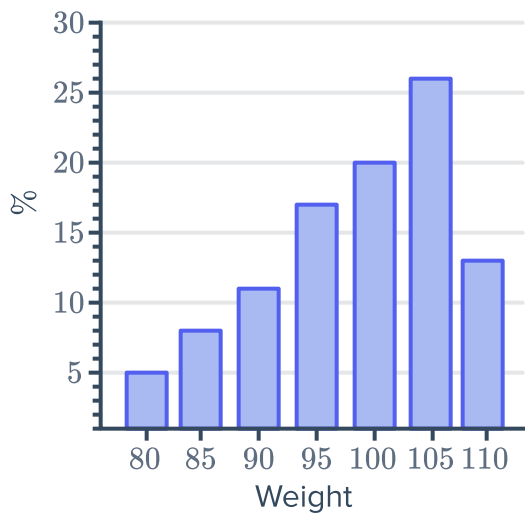
a



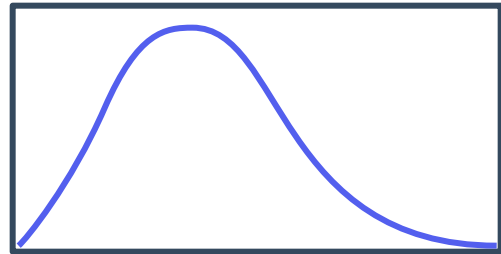
b



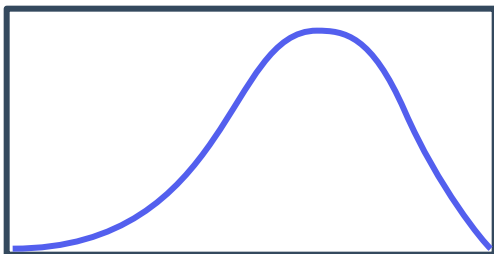
c



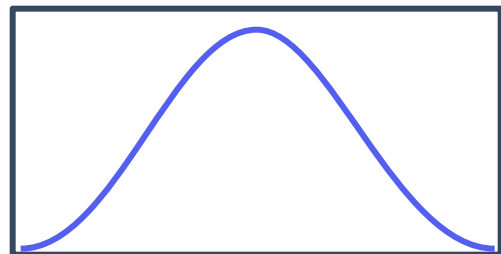
d



e



f



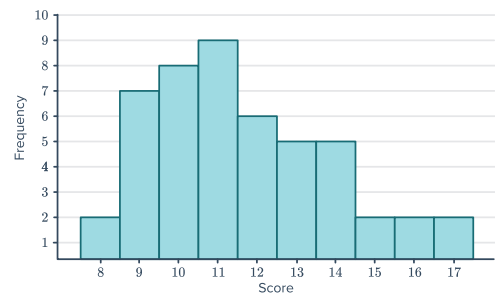
g



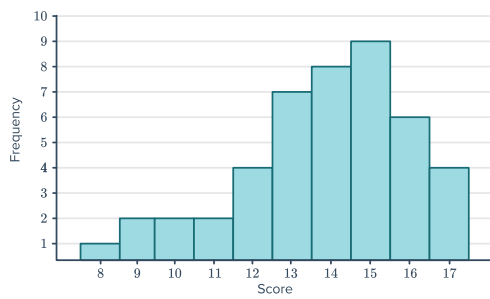
i



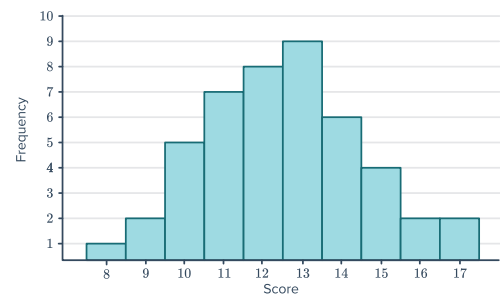
j



k



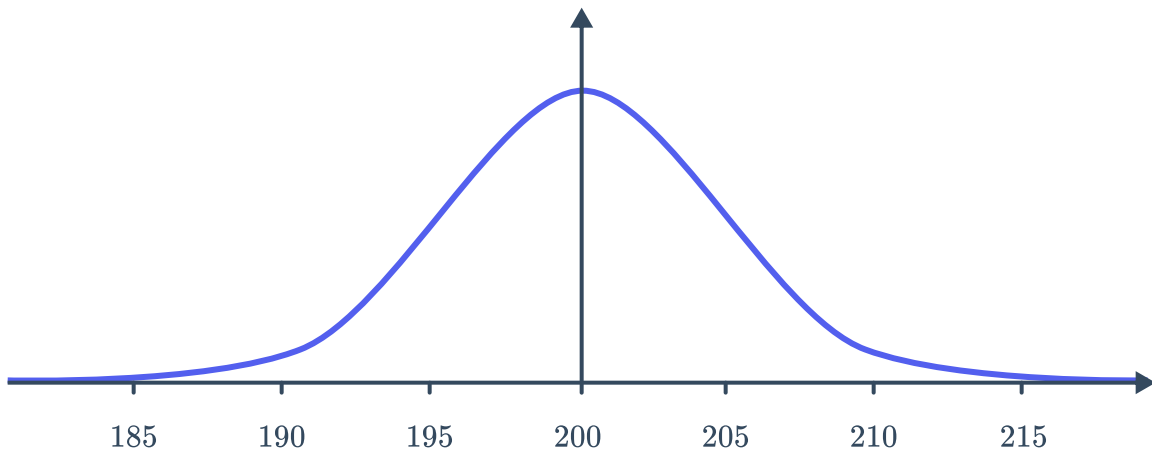
l



2 Consider the given normal distribution:



- a State the mean.
- b State the median.
- c State the mode.



3 State whether the following statements are true of the normal distribution curve:

- a There are the same number of scores above and below the mean.
- b The curve is symmetrical.
- c The fewest scores lie around the mean.
- d The curve is asymmetrical.
- e The spread of the normal distribution changes depending on the mean.
- f A greater mean will result in a skewed curve.
- g The mean, median and mode all have the same value.
- h The spread of the normal distribution changes depending on the standard deviation.

4 Given a normal distribution, state the percentage of people that lie:

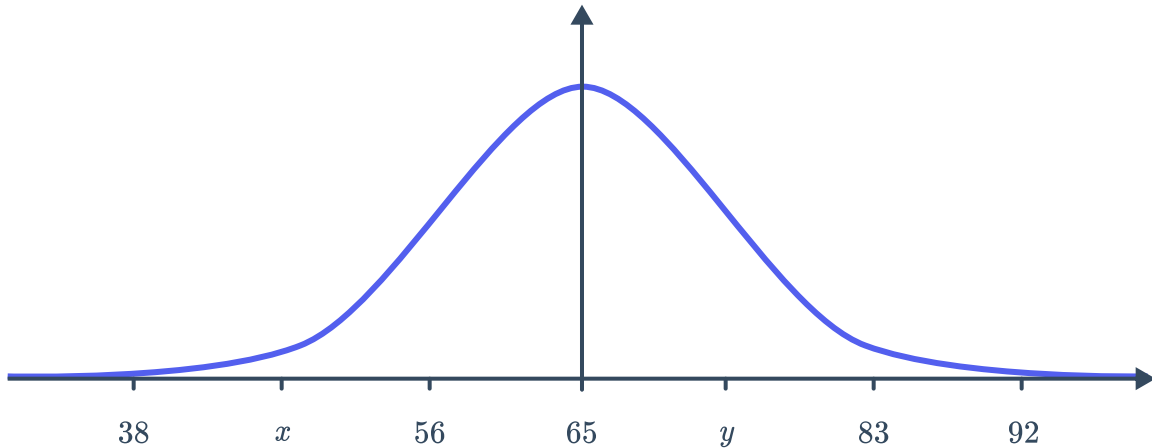
- a Above the mean.
- b Below the mean.

5 A data set has a mean of 80 and a standard deviation of 3. Find the score that is 3 standard deviations above the mean.



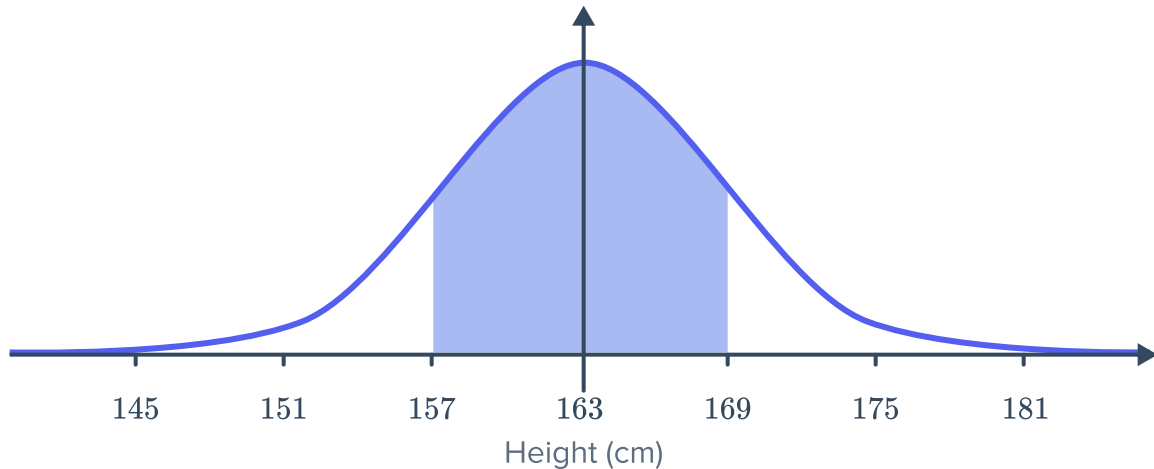
Applications

- 6 The results from an exam were approximately normally distributed. The mean score was 65, with a standard deviation of 9. The points marked on the horizontal axis are separated by 1 standard deviation:



- a State the value of x on the graph. b State the value of y on the graph.
- 7 In a male population, the mean height was 175 cm, with a standard deviation of 7 cm.
- a State the height of Bob who is 4 standard deviations above the mean.
- b State the height of John who is 2 standard deviations below the mean.
- 8 A sample of professional basketball players is normally distributed and gives the mean height as 199 cm with a standard deviation of 10 cm. State the height of a basketball player who is:
- a 3 standard deviations above the mean b 1.5 standard deviations below the mean

- 9 The diagram below shows the distribution of females heights in a population:



- a If 34% of females lie between 157 cm and 163 cm, what percentage of females lie between 163 cm and 169 cm.
 - b What length is one standard deviation equal to?
- 10 Assume the mass of sumo wrestlers is normally distributed, with a mean mass of 158 kg and a standard deviation of 10 kg.
- a What percentage of wrestler would weigh more than 158 kg?
 - b A sumo wrestler's weight is one standard deviation below the mean. How much does he weigh?